

TECHNICAL NOTES

DYNAMIC TRACE-GAS DILUTION SYSTEM

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Abstract—Apparatus has been developed for producing low gas concentrations in air samples of controlled characteristics. The gas diluting unit of the system consists of two Wösthoff rotating stopcock units placed in series and modified to eliminate concentration variation in output. The device can be used for all but the least stable gases, and will produce concentrations from 100 ppm to 0.1 ppb ($1:10^{10}$). Lower concentrations could be achieved if it became desirable.

INTRODUCTION

RECENT reviews by LODGE (1968) and SALTZMAN (1968) of current techniques for producing low gas concentrations in air samples demonstrate the need for a precision gas concentration calibration system of variable range. SALTZMAN and GILBERT (1959) prepared diluted gases by using a porous asbestos plug through which gas would diffuse. Because of variations in the packing of the plug and back pressure of the system, the apparatus had to be independently calibrated. Both the permeation tube gas dilution technique of O'KEEFE and ORTMAN (1966) and the Gas Dosing Device of H. Wösthoff, o. H.G.,* lack apparatus for purifying and metering the air itself to achieve a final diluted gas flow of accurately known humidity, purity, and gas volume.

Faced with the need of such a complete and precise gas dilution capability, a dynamic gas dilution system (FIG. 1) was constructed based on the accurate gas dispensing device of H. Wösthoff, o. H.G. This system can provide trace gas concentrations of 100 ppm to 0.1 ppb at flows up to 30 lpm.† This range is accessible using pure tank gas, and even lower concentrations could be achieved with commercial diluted tank gas. The gas dilution system seems to meet all requirements for use as a precision calibration device in the laboratory.

APPARATUS DESIGN

The following discussion traces the process of gas dilution through the apparatus. Parenthetical identifiers at each equipment stage refer to FIG. 1, a schematic diagram of the system.

Air under pressure enters the system through a Matheson Model-8 two stage regulator (R_1) (Matheson Co., Inc. East Rutherford, N.J.).‡ The second stage of the regulator is set at a constant 30 psig. Air leaving the regulator passes over solid purifying agents (CaSO_4 , activated charcoal, molecular sieve 5Å, and silica gel) and enters an ARO single stage regulator (R_2) (ARO Corp., Bryan, Ohio) set at a fixed 14 psig. A closed manometer-orifice flowmeter (M_1) calibrated at the 14 psig regulator pressure measures the dry air leaving the single stage regulator. (All manometer-orifice flowmeters used in the system are calibrated with a dry test meter, or for low flow rates, with a soap bubble flowmeter.)

A valve system diverts a specified portion of the air flow to water scrubbing towers and then recombines the flows. Valve V_1 controls the dry air; valves V_2 and V_3 control air diverted through flowmeter M_2 to the water scrubbing towers (T). Any relative humidity between zero and 95 per cent can be achieved by varying the fraction of the air passing through the towers.

The humidity-conditioned air now passes through a glass fiber filter (F) to remove water droplets,

* H. Wösthoff, o. H.G., Apparatebau, Bochum, Germany, distributed by Instrumentation Associates, Inc., 17 W. 60th St., New York, N.Y. 10023.

† ppm—parts per million, $1:10^6$; ppb—parts per billion, $1:10^9$.

‡ Designation of part numbers and manufacturers does not constitute an endorsement beyond the fact that these components function in the apparatus as described. Equivalent products, where they exist, could presumably be substituted.

and into a brass manifold. The manifold has several outlets, all fitted with quick disconnects (Hansen Mfg. Co., Cleveland, Ohio). Two valves (V_4 and V_5), each coupled with a flowmeter (M_4 and M_5 respectively), direct conditioned diluting air to the two stages of the Gas Dosing Device (W) (H. Wösthoff, o. H.G.).

This Gas Dosing unit consists of a mechanically stepped, precision bore stopcock which rotates 90° per impulse. At one position the stopcock bore fills with gas; at the next, prepared dilution air from the manifold is passed through the bore, mixing with and diluting the gas. A bypass allows for a

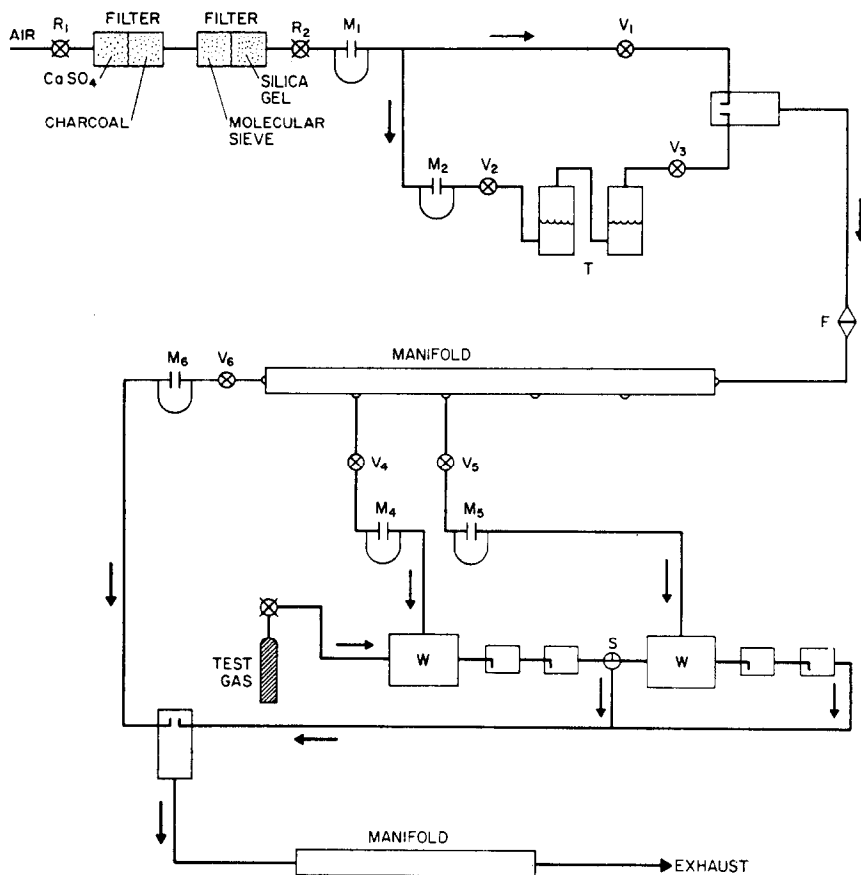


FIG. 1. Dynamic system for preparation of precise trace-gas concentrations. F, glass fiber filter (Millipore Corp.); M_1 , M_2 , M_6 , manometers (Meriam Instrument Co.); M_4 - M_5 , flowmeters (Pyrex T); R_1 , two-stage regulator (Matheson Co.); R_2 , single stage regulator (ARO Corp.); S, three-way stopcock (Teflon); T, glass fritted gas washing towers; V_1 - V_6 , 20-turn needle valves (Hoke, Inc.); W, gas dosing device, — M_2 (H. Wösthoff, o. H.G.).

continuous flow of the diluting air. An electrical impulse generator of variable frequency controls the stepping rate, and hence the dilution. Given the volume of the bore, the stepping rate of the stopcock, and the flow rate of diluting air, the dilution may be computed precisely. A second identical gas dosing unit used in series with the first will produce a maximum dilution of $1 : 10^8$.

The air/gas mixture from the dosing unit can be further diluted with a constant high volume flow of air from the manifold via a third outlet valve (V_6) with flowmeter M_6 . Varying the air flow rates of the Wösthoff units and the rate at which the stopcock is filled with gas and emptied yields a wide range of gas concentrations at a nearly constant output flow rate.

Equilibration time of the clean walls of the apparatus is about 3h. A 2h wait is required for

each change in concentration during a run, the time being dependent upon the wall materials and the system's total surface area. Reproducibility of the gas flows, as depicted by the manometer calibration curves, is within 2 per cent when the valves V_1 and V_2 are the principal flow controls. These needle valves will maintain relatively constant pressure across the manometers, and the slight pressure drops in the rest of the system will have little effect on the calibration curves.

RESULTS

The diluting system was tested by setting the apparatus for a particular concentration of test gas and then analyzing for it. The gases tried were H_2S , analyzed by the method of AXELROD, CARY, BONELLI, and LODGE (1969) and SO_2 using the procedure of WEST and GAEKE (1956). TABLE 1 gives

TABLE 1. DOUBLE-DILUTION COMPARISON STUDY

Concentration required of Wösthoff unit	Concentration found
100 ppb* SO_2	93.6 ppb* SO_2
50	50.8
10 ppb* H_2S	11.2 ppb* H_2S
3.0	2.8
0.80	0.82
0.20	0.24

* ppb = $1:10^6$ dilution.

the results of these tests. (STRATMANN and BUCK (1966) used the basic dosing device to produce diluted NO_2 in the ppm range.)

CONCENTRATION VARIATION OF OUTPUT

A major disadvantage of the Wösthoff dilution unit, fluctuating concentration in output caused by the sudden pulsing emission of gas into the dilution stream, was pointed out by TOKIWA and DEVERA (1967). To eliminate this, a series of flasks was placed after the Wösthoff units, progressively diminishing the variation in the concentration of the gas pollutant.

Consideration of such a series of flasks leads to the equation:

$$\frac{dC_{it}}{dt} = \left[\frac{C_{(i-1)t} - C_{it}}{V_i} \right] A,$$

which describes the behavior of the system, where C_{it} is the concentration of test gas leaving i th flask at time t , $C_{(i-1)t}$ is the same for the $(i-1)$ st flask, V_i is the volume of the i th flask, and A is the volume air flow through the flask.

This equation was analyzed graphically by computer for a number of values of air flow (50–400 $ml \cdot min^{-1}$) and flask volume (100–2000 ml), with the number of flasks varying from one to four, and with a square-wave input ($1-10 \cdot min^{-1}$) to the first flask. Computer output described the variation of concentration in each flask at steady state, and the time required for the flask to reach steady state. As might be expected, the computer analysis proved that several small flasks are better than one large flask. Three 500 ml flasks in series after the Wösthoff units proved suitable, reducing final theoretical oscillations in concentration of 2 per cent. In a test experiment, a Mast Ozone monitor (sensitive to NO_2) showed oscillation of roughly 25 per cent in a nominal 20 ppm concentration of NO_2 using a single Wösthoff unit with no flasks. No oscillation was evident in the final gas flow after three 500 ml flasks were added.

DISCUSSION

This gas dilution apparatus has several advantages over permeation tubes for producing calibrated low gas concentrations to be used as laboratory standards. Permeation tubes are inexpensive, lightweight, and convenient for field use, but they have serious limitations. Only liquefiable gases can be used with the permeation tubes, and gases such as CO₂ and NO₂ which affect Teflon can weaken and rupture the walls of the tube. Tubes with oxygen-reactive and water-reactive contents must be stored under special conditions. The tubes themselves are simple, but the equipment in which they are used requires that air flow and temperature be maintained at 1 per cent and 0.1 per cent accuracy respectively. For precision calibration by weighing, these conditions must be sustained for 15–30 days. The output flow of the permeation tube system is determined by the desired gas concentration, unless special plumbing arrangements are made for secondary dilution. Such a rigidly determined flow can cause sampling problems.

The modified Wösthoff system has no such limitations. It can accept all gases, and does not demand stringent environmental controls. Final concentration can be selected and varied at the will of the operator, with a large or small constant flow maintained for parallel or special sampling purposes. The system described provided for dilution of a single gas, but with more dosing units and additional plumbing dilute gas mixtures can be obtained. The Wösthoff system has great versatility and utility for the preparation of highly dilute gas standards in the laboratory.

The authors wish to note the contribution of David W. Fulker, who derived the theoretical equation and the computer program used.

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REFERENCES

- AXELROD H.D., CARY J.H., BONELLI J.E., and LODGE J.P., JR. (1969) "Fluorescent determination of sub-ppb H₂S in the atmosphere." *Analyt. Chem.* **41**, 1856.
- LODGE, J.P., JR. (1968) Production of controlled test atmospheres. In *Air Pollution* (edited by A.C. Stern) Chap. 27, Vol. II, 2nd ed. Academic Press, New York.
- O'KEEFE A.E. and ORTMAN G.C. (1966) Primary standards for trace gas analysis. *Analyt. Chem.* **38**, 760.
- SALTZMAN B.E. (1968) Air quality program needs uniform tests. *Environ. Sci. Tech.* **2**, 22.
- SALTZMAN B.E. and GILBERT N. (1959) Microdetermination of ozone in smog mixtures: nitrogen dioxide equivalent method. *Am. Ind. Hyg. Ass. J.* **20**, 379.
- STRATMANN H. and BUCK M. (1966) Messung von Stickstoffdioxid in der Atmosphäre. *Int. J. Air Wat. Pollut.* **10**, 313.
- TOKIWA Y. and DEVERA E.R. (1967) The Wösthoff gas dosing apparatus. Presented at the *Eighth Conference on Methods in Air Pollution and Industrial Hygiene Studies*, Oakland, California 6–8 February.
- WEST P.W. and GAEKE G.C., JR. (1956) Fixation of sulfur dioxide as disulfitomercurate (II) and subsequent colorimetric estimation. *Analyt. Chem.* **28**, 1816.

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